

NATORI-ON-PSYFR

The Predictive Chronology Engine

TECHNICAL REFERENCE

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github.com/BeeRadicalStuff/natori-on-psyfr

beeradicalstuff.github.io/natori-on-psyfr (live)

A worldbuilding & study instrument rendering the ARCHAIX thesis of Jason Breshears as living predictive calendrics. Presented as that thesis — not as established science.

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1. Purpose & Lineage

NATORI-ON-PSYFR fuses two instruments. The first is the **Ophis prediction grammar** — reverse-engineered from the Ophis v12 browser tool — in which two known dates (**X-Dates**) define an interval **Y** measured in axial rotations (days), and a library of arithmetic **operations** transforms that interval into projected **Z-Dates**. The second is the **Chronicon calendrics** — the Breshears cycle lattice (Phoenix 138 · Nemesis 792 · NER 600 · Metonic 19), Annus Mundi dating, the Mayan Long-Count, and the šar reckoned as a turning of the stars (a day).

The fusion — and the part that is new in this project — is the **scoring**: every projected date is weighed against the documented chronology and the signature numbers, so the instrument itself reports when a projection lands on a Phoenix node, a documented event, a palindrome, an eclipse, or a resonance number.

1.1 Deliverables

File	Role
clocks/PSYFR1.html	The engine (main page). Single-file, offline, no network calls. Distributed as NATORI-ON-PSYFR-Main-UI.html.
clocks/PSYFR2.html	The field guide: quick start, formula cookbook, grammar, tag key, worked examples, FAQ. Distributed as NATORI-ON-PSYFR-Field-Guide.html.
test/verify.js	Math regression harness — chronology anchors, calendar round-trips, grammar, the 2040 convergence.
test/runtime.js	Boot harness — runs the full engine script under a fake DOM to catch load-order bugs.
docs/...Walkthrough.docx/.pdf	Teaching write-up of the build.
CLAUDE.md / README.md	Project context for Claude Code sessions; human quick-start for the repo.

1.2 The contract between the two pages

The two HTML files cross-link **by bare filename** and therefore must live in the same folder. They share the localStorage keys `ophion-theme` and `ophion-zoom` (theme and text size carry across both). The engine alone also stores `ophion-mode` — the ♦ Simple view toggle. These three UI-preference keys are the only browser storage the app uses; anchors and operations are deliberately never persisted.

2. Architecture

The engine is one self-contained HTML file: CSS, markup, data tables, and a single inline script. There is no build step and no external request other than optional Google Fonts (the page degrades gracefully to system serif when offline). The script is organised in banners:

- **OPHION CORE** — verified date and prediction math: Julian Day conversion, chronology offsets, the `oph_*` functions, MSRF set, eclipse tables.
- **PREDICTION ENGINE** — operation compiler, scoring, the cast, convergence clustering.
- **STATE + UI** — default operation packs, ledger data, rendering, event wiring, theme/zoom/mode persistence. (The verify harness extracts everything before this banner as the pure-math core, so the banner text is load-bearing.)

The page presents five sections: **The Oracle** (seed anchors → choose operations → cast → scored Z-Dates), **The Convergence** (where independent operations agree), **The Wheels** (one moment read across every reckoning), **The Ledger** (the documented event record), and **Method** (scoring rules and framing).

3. The Calendar Core

All dates are handled as **astronomical years** (year 0 = 1 BC, -1 = 2 BC ...) over the proleptic Gregorian calendar, converted to and from **Julian Day Numbers** by the standard integer algorithms:

```
function jdn(ay, m, d)    // astronomical y/m/d -> Julian Day Number
function jdToDate(J)      // Julian Day Number -> {year, month, day}
fmtYear(a)               // 0 or less prints as BC: fmtYear(-2238) = "2239 BC"
```

Every chronology is an offset or a modulus over that JDN spine:

Reckoning	Rule	Notes
Annus Mundi	AM = ay + 3894	Epoch jdn(-3894, 1, 1); world-year count.
Long-Count year	LC = ay + 3112	Mayan correlation; LC 5138 = 2026 CE.
Since Cataclysm	ay + 5238	Years since the Nemesis Cataclysm (5239 BC).
Phoenix lattice	ay mod 138 = 108	Every documented Phoenix event sits exactly on this lattice.
Nemesis X	792-yr cycle	Inner-transit window flagged separately.
Anunnaki NER	600-yr cycle	NER node when offset = 0.
Baktun edges	MAY_NODES list	14 hard-coded baktun years from -3112 to 2046.
Metonic moon	19 yr = 235 moons	Synodic month 29.530588853 d from the J2000 new-moon epoch.

3.1 Sanity anchors

These fixed points are asserted by the test harness and were verified against the Chronicon before any UI work began:

Assertion	Value
Today (2026 CE)	AM 5920 · LC year 5138
The Great Flood (astro -2238 = 2239 BC)	AM 1656
Petrie's survey year (1882 CE)	AM 5776
All 24 Phoenix events	mod 138 = 108, exactly

4. The Operation Grammar

An operation is a string. It must start x1+ or x2+ (the earlier / later anchor of the pair), followed by arithmetic on **Y** — the interval in days. The compiler, `compileOp(eqRaw)`, enforces the grammar in five steps:

- Normalise: strip whitespace; a bare x or × between terms becomes * (multiply).
- Require the x1+ / x2+ prefix; everything after it is the offset body.

- Whitelist check: after substituting the known tokens away, any remaining character outside 0-9 + - * / () . rejects the equation ("illegal token").
- Compile with new Function(...) — a syntax error rejects it.
- Probe with Y = 1000 — if the result is not a finite number, reject ("does not evaluate to a number").

Token	Meaning
Y	Interval between the anchor pair, in days (axial rotations).
OPH_PHI	Golden ratio, 1.61803398875.
OPH_PI	Archaix π , 3.14 (deliberately the lore value, not Math.PI).
OPH_CRV	Curvature constant, 5.08.
oph_flip(n)	Digit reversal: 138 → 831, 19 → 91. Decimal points keep their position.
oph_round / floor / ceil	Round to a whole day (nearest / down / up).
oph_abs / oph_sqrt	Absolute value; square root of the absolute value.

5. The Cast Pipeline

cast(anchors, ops, system) walks every unordered pair of enabled anchors and every enabled operation:

```
for each anchor pair (x1 earlier, x2 later):
    Y = |x2.jd - x1.jd|           // days; skip if Y < 1 or Y > 3,000,000
    for each operation:
        offset = op.fn(Y)         // skip non-finite
        ZJD    = round(base + offset) // base = x1.jd or x2.jd per the op
        skip if projected year < -5400 or > 4000
        dedup on (ZJD, equation, pair)
        score = scoreZDate(...) + MSRF bonus + eclipse bonus
        tag ECHO if ZJD lands within ±1 day of ANY input anchor
```

Results sort by score (descending), then by date. An **ECHO** is a projection that merely re-derives one of the inputs — it is kept, dimmed and tagged, because it confirms the cast is internally consistent, but the ♦ NOVEL ONLY filter exists to hide echoes and surface only dates the chronology did not already name.

6. Scoring — the Two Lenses

Every Z-Date is scored by scoreZDate(year, JD, system). The same checks run under both lenses; only the weights change. **V8 (chronology-first)** asks "which projections resonate with history?" — **V7 (numbers-first)** is the legacy Ophis-flavoured lens where the sacred numbers dominate. Switching the lens re-scores and re-sorts the table live.

Criterion	V8	V7	Test
Phoenix node	+5	+2	year mod 138 = 108
Near Phoenix (±2 yr)	+2	+1	distance to node ≤ 2
Documented event	+5	+1	year in ledger set
Palindrome ↵	+3	+4	year, AM or JD reads both ways
138 in the digits	+2	+4	"138" in year / AM / LC

Criterion	V8	V7	Test
Metonic-19 to today	+2	+2	(today - year) mod 19 = 0
19 in the digits	+1	+2	"19" in year / AM
MSRF resonance	+2	+3	Y or offset in MSRF set
Solar eclipse ±1 day	+2	+2	precomputed table hit
Lunar eclipse ±1 day	+1	+1	precomputed table hit
Nemesis / NER / baktun	+1 each	+1 each	cycle membership

The **MSRF set** is the Ophis resonance table (key NORMAL + IMPORTANT members: 12, 21, 24, 36 ... 138 ... 414, 552, 600, 792, 828, 831 ... 1656), with **19 added by owner request** — it is not in the stock Ophis table.

7. Convergence Detection

A single high-scoring projection is a guess; the ARCHAIX signal is **agreement**.

`findConvergences(results, tol)` clusters projections that land within a window — exact day, ±1, ±7, ±30 (default), ±90 days, or same calendar year — by sorting on JD and chaining any neighbour within tolerance. Each cluster reports:

- **Strength** — the number of distinct operations that agree (clusters below 2 are discarded).
- How many distinct **anchor pairs** contributed — agreement across different pairs is stronger than one pair repeating itself.
- The centre date (mean JD), the span in days, the best member score, and the union of all non-echo tags.

Clusters sort by strength, then best score, then tightness. A cluster whose members are all echoes is flagged and excluded from the headline list.

8. The Eclipse Tables

Real solar and lunar eclipse dates (~1–3000 CE, from the Ophis precomputed data) ship inside the file as **delta-encoded strings**: a base JD plus comma-separated day-gaps, with a parallel type string (T total, A annular, P partial, H hybrid). At boot they decode into sorted JD arrays; `eclipseNear(jd, tol)` answers by binary search. Projections outside the covered range simply carry no eclipse tag — the instrument degrades gracefully rather than guessing.

9. The Ledger & The Wheels

The ledger is a 69-event array of tuples [`astroYear`, `kind`, `text`, [`month`, `day`]] — the dated spine of the thesis from the Nemesis Cataclysm (5239 BC) through the 2046 close of the count. Kinds: key, phoenix, nemesis, ner, baktun, note. Every phoenix entry was verified to sit on the 138-lattice before release. Any row can be **seeded** as an anchor (it loads with its month/day) or clicked to read on the wheels.

The Wheels re-read one moment across seven reckonings: Phoenix (138 yr), Nemesis X (792 yr), Anunnaki NER (600 yr), Annus Mundi, Mayan Long-Count, Anunna Turnings (šar = a day), and the Metonic moon (19 yr = 235 lunations). A dial (year/era/month/day plus a slider from 2843 BC to 2178 CE) drives them directly; preset jumps go to Today, 2040, 2046, and 2178.

10. The Default Pack & Themed Packs

Nineteen operations load by default — the palindromic cast:

Operation	Reading
$X2 + \text{oph_round}(Y)$	The Isometric Date — the same interval, forward again.
$X2 + \text{oph_flip}(\text{oph_round}(Y))$	The Holo-mirror — reverse the digits, project from the later anchor.
$X1 + \text{oph_flip}(\text{oph_round}(Y))$	The same mirror from the earlier anchor.
$X2 + Y / \text{OPH_PHI}$	Golden division.
$X1 + Y * \text{OPH_PHI}$	Golden expansion.
$X1 + (Y/2) * \text{OPH_PI}$	The pi half-arc.
$X2 + Y / \text{OPH_CRV}$	Curvature division (5.08).
$X2 + Y * 138 / 100$	Scale by 1.38 — the Phoenix stretch.
$X1 + Y * 19 / 10$	Scale by 1.9 — the Metonic stretch.
$X2 + \text{oph_round}(Y/138) * 138$	Snap to the 138-grid.
$X1 + \text{oph_round}(Y/19) * 19$	Snap to the 19-grid.
$X2 + Y + 138$	One Phoenix count beyond the isometric date.
$X1 + Y + 19$	One Metonic seal beyond it.
$X2 + \text{oph_flip}(Y) + 19$	Mirror, then seal with 19.
$X1 + Y * 360 / 365.2422$	Reframe into the 360-day Draconian year.
$X2 + Y * 792 / 600$	Nemesis-to-NER ratio.
$X1 + \text{oph_round}(Y * \text{OPH_PHI} / \text{OPH_PI})$	ϕ over π .
$X2 + \text{oph_round}(Y / \text{OPH_PHI} / \text{OPH_PHI})$	Divide by ϕ^2 — the tighter golden step.
$X1 + \text{oph_flip}(\text{oph_round}(Y / \text{OPH_PHI}))$	The golden mirror — divide by ϕ , then reflect.

Four themed packs can replace the default set with one click: the **138 Pack** (12 ops on the Phoenix number and its multiples 414/552), the **19 Metonic Pack** (11 ops on 19/235), the **Phoenix Lattice Pack** (10 cycle-snaps across 138/792/600/360/216/144), and the **Golden Pack** (10 ops on ϕ , π , and the curvature constant). Custom operations typed into the field join the active set after passing the compiler.

10.1 Configuration files

↓ Save .json writes the whole cast — anchors, operations, scoring lens, convergence window — as {app:"OPHION", v:3, ...}. ↑ Load restores one. The files are plain JSON and freely shareable; this is the same portable-model idea as the original Ophis .oph files.

11. The UI Layer

The aesthetic is Cinzel / EB Garamond / IBM Plex Mono — gold #d8a943, red #d3402f, cyan and violet on near-black, with an aged-parchment light theme. Both pages honour prefers-color-scheme on first visit and prefers-reduced-motion always; every control is keyboard-reachable with a visible focus ring, and A− / A / A+ scales the whole page (0.8–1.5×).

11.1 Progressive disclosure (v1.0)

The v1.0 declutter keeps every capability but folds the heavy panels behind one-click expanders, closed by default: the 19-operation list, the op-packs / save-load toolbar, the 69-row ledger table, and the scoring rulebook. A numbered Seed → Cast → Read strip replaces the dense intro. The thesis-framing card in Method deliberately stays un-foldable.

11.2 Simple view

The ♦ Simple toggle (top-right, persisted as ophion-mode) reduces the page to the essentials — anchors, the cast button, results, convergence, and the wheels — by hiding the moment bar, the operations panel, the scoring lens, the advanced fold, the Ledger, and Method (their nav links hide too). It is a pure CSS class on body; the engine underneath is untouched, so a cast in Simple view produces identical results. ☼ Full view restores everything.

12. Verification — the Discipline

The house rule is verify-first: prove the math before touching UI, and after any change to the engine run both harnesses (npm test).

Harness	What it proves
test/verify.js	11 checks: today → AM 5920 / LC 5138; Flood → AM 1656; Petrie → AM 5776; oph_flip(138) = 831 and (19) = 91; isPalindrome(1331); 3,276 calendar round-trips; all 81 shipped operations compile; the grammar rejects 3 known-bad strings; the default cast yields the strength-4 convergence on 2040. It extracts the pure-math core from the HTML by cutting at the STATE + UI banner.
test/runtime.js	Boots the entire inline script under a minimal fake DOM (Proxy-based element stubs, localStorage shim). Catches load-order and temporal-dead-zone bugs that a parse check cannot — "it parses" is not "it runs".

Both harnesses passed 11/11 + clean boot before and after every v1.0 change, and the live page was additionally smoke-tested in a real browser: default cast = 34 projections, 5 convergences, top result 10 May 2040 (score 12), zero console errors.

13. Release & Distribution (v1.0)

- **Git tag** v1.0 — commit 7521518 on main; the exact released state is always recoverable.
- **GitHub Release** — github.com/BeeRadicalStuff/natori-on-psyfr/releases/tag/v1.0 with the distributable zip attached.
- **Live page** — GitHub Pages serves the repo root; index.html redirects into clocks/PSYFR1.html. Also embedded on the Wix site as an iframe, which therefore tracks the live build automatically.
- **The zip** — NATORI-ON-PSYFR-v1.0.zip: README.txt (quick start, three worked examples, paste-ready formulas, troubleshooting), the two HTML files under their friendly names with cross-links rewritten, and docs/ with the Technical Walkthrough in PDF and DOCX.

13.1 Renaming rule

Distribution copies rename the files but **never the internals**: visible text, cross-links and download filenames may change; function names, element IDs and storage keys must not. The packaging step is two string replacements — PSYFR2.html → NATORI-ON-PSYFR-Field-Guide.html inside the engine and PSYFR1.html → NATORI-ON-PSYFR-Main-UI.html inside the guide.

14. Known Limits & Next Steps

- Eclipse coverage is ~1–3000 CE; outside it the tag is silently omitted. Widening the tables is a known next step.
- Deep-past conversions are cycle-true rather than historically observed — stated in-app.
- Intervals above 3,000,000 days and projections outside 5400 BC – 4000 CE are skipped by design.
- The single-file form is a feature (open-from-disk, air-gapped); splitting into modules remains an option but must not break that property.
- Candidates: a printable cheat-sheet, engine screenshots embedded in the walkthrough, more themed packs.

NATORI-ON-PSYFR v1.0 · Ophis grammar × Chronicon calendrics over the Breshears lattice. Self-contained & air-gap friendly — nothing leaves the page. Presented as the ARCHAIX thesis of Jason Breshears (Archaix.com), not as established science.